

Kuan Yi Wang

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EDUCATION

University of Waterloo

Bachelor of Computer Science | Coursework: Data Structures & Algorithms, Calculus I & II

Expected April 2029

Waterloo, ON

TECHNICAL SKILLS

Languages & Data: Python, C/C++, SQL, Bash, NumPy, pandas, scikit-learn

Deep Learning & Robotics: PyTorch, MuJoCo, nnU-Net, CUDA, DDP, VLM/VLA, world modeling, BF16 mixed precision

Engineering: Git, Linux, Conda, Virtual Environments (venv), CI/CD | Codex, Claude Code, Cursor

EXPERIENCE

RBC Borealis

Sept. 2026 – Dec. 2026

Machine Learning Fellow

- Proposed OpenLatent, using adversarial representation learning to train on encoded wearable medical data without sharing raw records; planned UCI HAR benchmarks for activity recognition, identity leakage, and reconstruction attacks.

Morgan Stanley

May 2026 - Aug. 2026

Software Engineer Intern

Montréal, QC

- Built Python automation for parallel AI agents to implement trading-rule changes across SQL schemas; accelerated rule insertion 20x for a Morgan Stanley brokerage platform (Pantheon) processing billions of transactions daily.
- Orchestrated Jira intake, code changes, SQL updates, and unit/regression testing through a Python scripting CI/CD workflow.

Neuropoly (Mila)

June 2025 - Jan. 2026

Data Scientist Intern

Montréal, QC

- Improved segmentation accuracy by 45% by training and fine-tuning PyTorch 3D U-Nets on MRI scans to identify spinal rootlets and multiple sclerosis lesions, developing medical-imaging models for the open-source Spinal Cord Toolbox.
- Accelerated 3D U-Net inference by over 125% through MRI cropping and resizing focused on spinal anatomy.
- Accelerated MRI/CT annotation and segmentation by over 80% through contributions to the SlicerCART image viewer.

PROJECTS

Pi 0.7 | Vision-Language-Action for Robot Manipulation | [GitHub](#)

Oct. 2026

- Recreated Pi 0.7 in PyTorch and fine-tuned it to move objects following spoken instructions; combined camera views, language, and robot joint states to predict continuous arm movements and gripper actions.
- Built a MuJoCo testbed with 2 SO-101 arms and 4 camera views; implemented a data pipeline for 930 demonstrations, aligning video, joint states, and 50-step action targets with episode-level train/validation splits.

Graph Neural Network for Quantum Error Correction | [ISEF Project Gallery](#) · [GitHub](#)

Oct. 2025 - May 2026

- Built a PyTorch Geometric GraphSAGE decoder to identify quantum-computing errors; used 80% fewer parameters than a dense neural network at distance 7, with 94.8% versus 89.3% accuracy on 10K simulated samples.
- Benchmarked graph decoders across code sizes and noise levels; averaged 23K samples/second over five timing runs on an RTX 5070 Ti at distance 7 and batch size 64, measuring decoding accuracy and batched inference throughput.
- Earned a Silver Award at the International Science and Engineering Fair (ISEF), Physics & Astronomy.

Genie-Inspired Doom World Model | Generative Video | [Paper](#) · [GitHub](#)

Sept. 2026

- Trained a PyTorch VQ-VAE on 600K Doom gameplay frames using CUDA and distributed training on 2 H100 GPUs; achieved 37.43 dB reconstruction PSNR across 4,096 frames from 256 test clips for a three-stage world model.
- Implemented spatiotemporal Transformers and latent-action learning for video prediction; reduced MaskGIT masked-token validation loss from 7.10 to 1.94 across 128 clips and published an illustrated write-up of the architecture and experiments.

LEADERSHIP

Global Talent Forum (Piñata Pitch)

Sept. 2024 - Present

Co-Founder

San Francisco Bay Area, CA

- Co-founded the largest student tech pitch competition based in Montreal (2000+ members), [a16z Tech Week member](#).
- Organized [AI Build Day x Pinata Pitch](#) hackathon with Shopify, helping more than 200 builders launch their startup.
- Raised \$50 000 and organized Piñata Pitch x Startupfest 2026, backed by Mila, Botpress and Morgan Stanley.